

SCREENSHOTS OF WORKING OF THE PROJECT

Done by:

YUVANESH.R

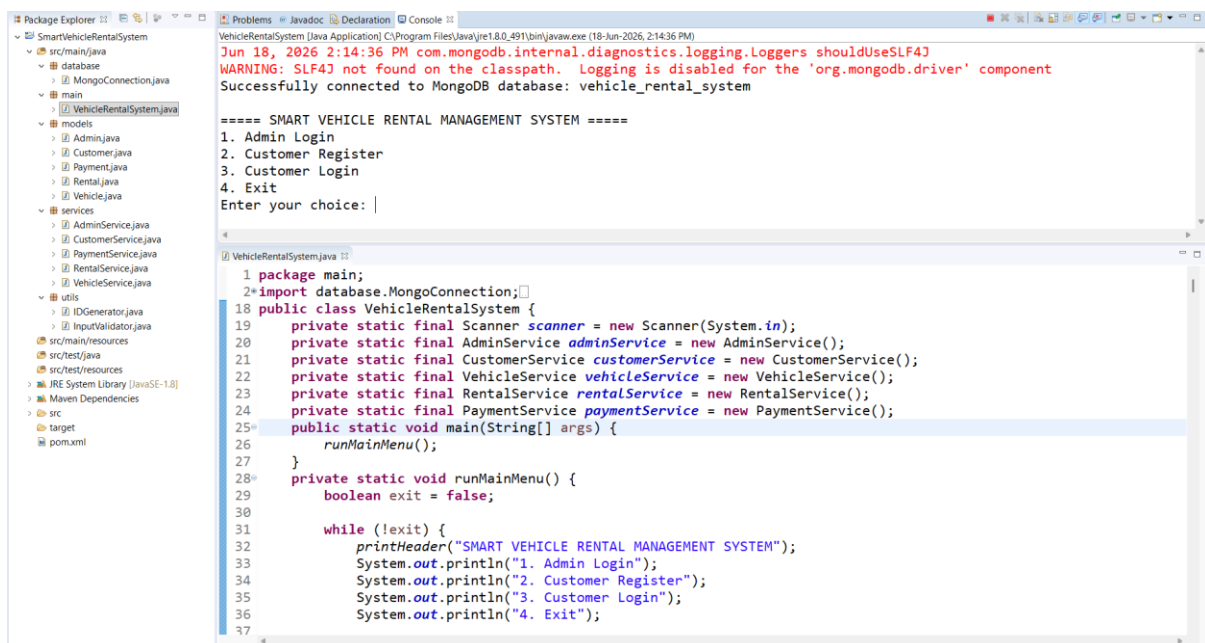
24BCE2487

Guided by:

JAYANTH YUVA

FACE

SAMPLE-1:

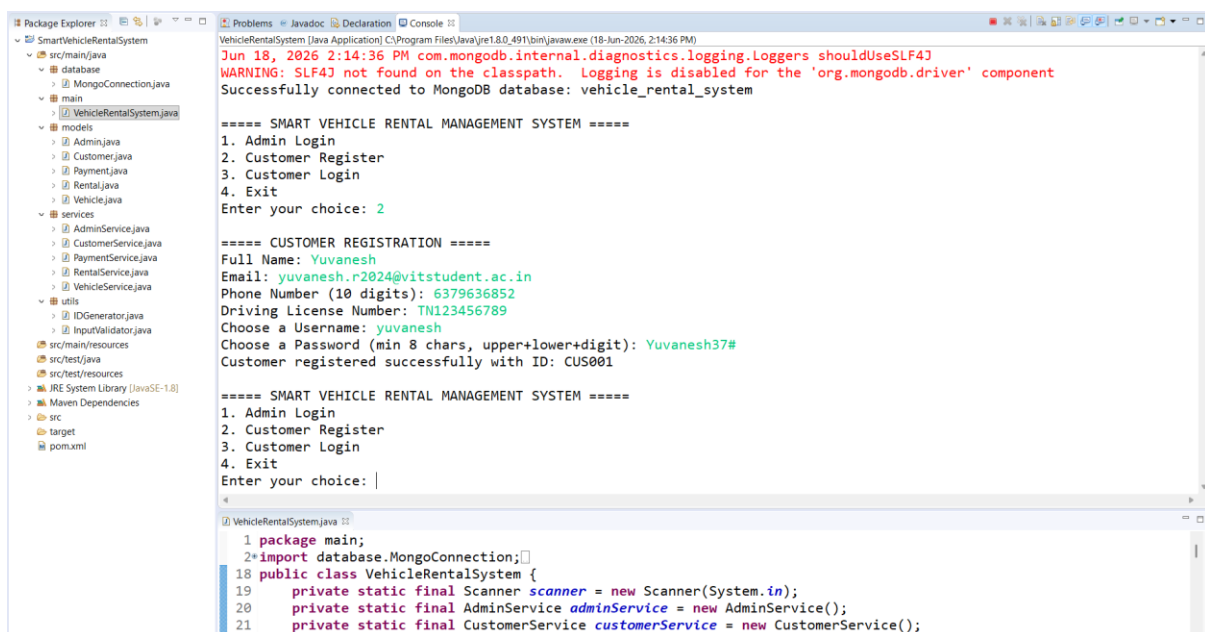


```
Package Explorer | Problems | Javadoc | Declaration | Console
SmartVehicleRentalSystem
  src/main/java
    database
      > MongoDBConnection.java
    main
      > VehicleRentalSystem.java
    models
      > Admin.java
      > Customer.java
      > Payment.java
      > Rental.java
      > Vehicle.java
    services
      > AdminService.java
      > CustomerService.java
      > PaymentService.java
      > RentalService.java
      > VehicleService.java
    utils
      > IDGenerator.java
      > InputValidator.java
  src/main/resources
  src/test/java
  src/test/resources
  JRE System Library [JavaSE-1.8]
  Maven Dependencies
  target
  pom.xml

VehicleRentalSystem [Java Application] C:\Program Files\Java\jre1.8.0_491\bin\javaw.exe (18-Jun-2026, 2:14:36 PM)
Jun 18, 2026 2:14:36 PM com.mongodb.internal.diagnostics.logging.Loggers shouldUseSLF4J
WARNING: SLF4J not found on the classpath. Logging is disabled for the 'org.mongodb.driver' component
Successfully connected to MongoDB database: vehicle_rental_system

===== SMART VEHICLE RENTAL MANAGEMENT SYSTEM =====
1. Admin Login
2. Customer Register
3. Customer Login
4. Exit
Enter your choice: |

VehicleRentalSystem.java
1 package main;
2 import database.MongoConnection;
18 public class VehicleRentalSystem {
19     private static final Scanner scanner = new Scanner(System.in);
20     private static final AdminService adminService = new AdminService();
21     private static final CustomerService customerService = new CustomerService();
22     private static final VehicleService vehicleService = new VehicleService();
23     private static final RentalService rentalService = new RentalService();
24     private static final PaymentService paymentService = new PaymentService();
25     public static void main(String[] args) {
26         runMainMenu();
27     }
28     private static void runMainMenu() {
29         boolean exit = false;
30
31         while (!exit) {
32             printHeader("SMART VEHICLE RENTAL MANAGEMENT SYSTEM");
33             System.out.println("1. Admin Login");
34             System.out.println("2. Customer Register");
35             System.out.println("3. Customer Login");
36             System.out.println("4. Exit");
37         }
38     }
39 }
```



```
Package Explorer | Problems | Javadoc | Declaration | Console
SmartVehicleRentalSystem
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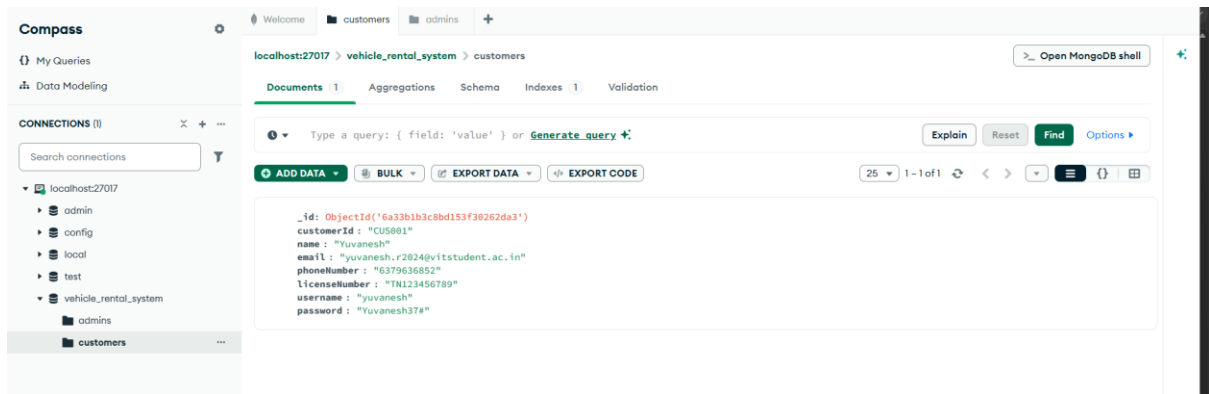
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===== SMART VEHICLE RENTAL MANAGEMENT SYSTEM =====
1. Admin Login
2. Customer Register
3. Customer Login
4. Exit
Enter your choice: 2

===== CUSTOMER REGISTRATION =====
Full Name: Yuvanesh
Email: yuvanesh.r2024@vitstudent.ac.in
Phone Number (10 digits): 6379636852
Driving License Number: TN123456789
Choose a Username: yuvanesh
Choose a Password (min 8 chars, upper+lower+digit): Yuvanesh37#
Customer registered successfully with ID: CUS001

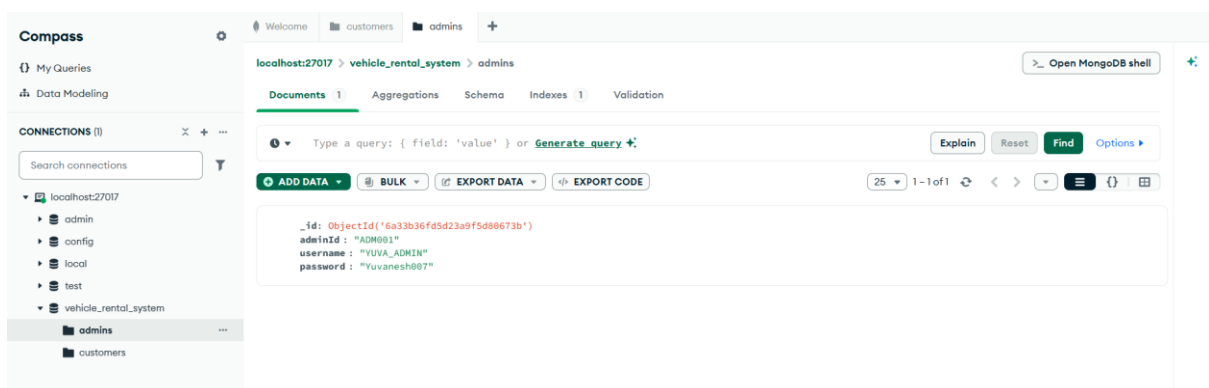
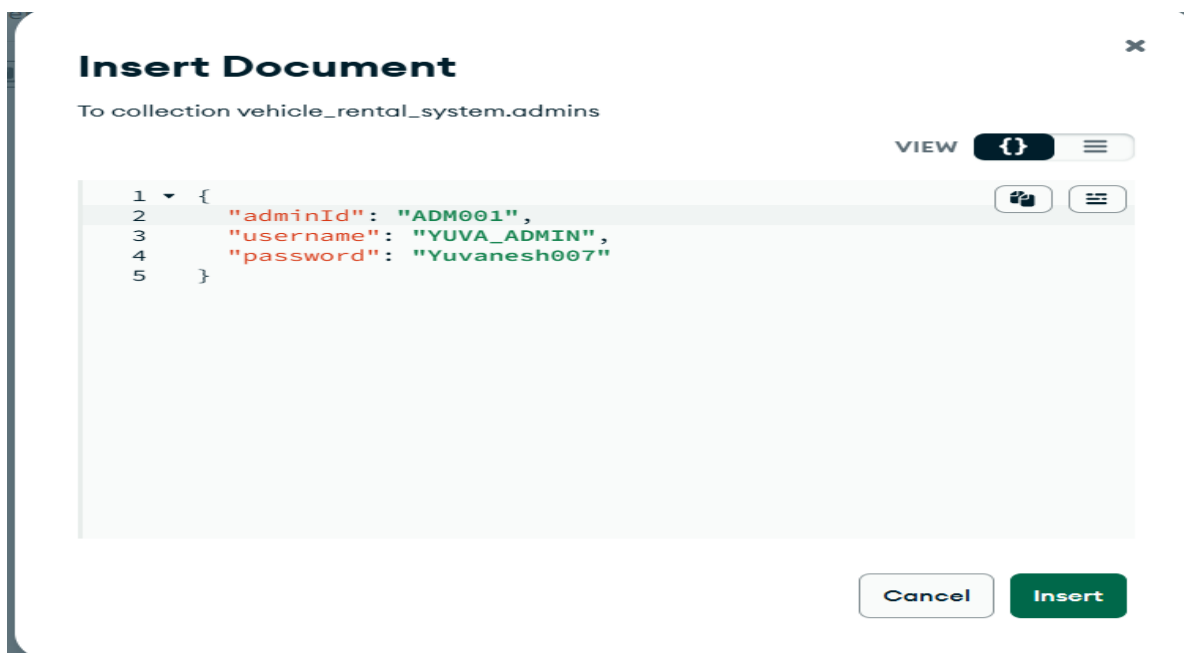
===== SMART VEHICLE RENTAL MANAGEMENT SYSTEM =====
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4. Exit
Enter your choice: |

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23     private static final RentalService rentalService = new RentalService();
24     private static final PaymentService paymentService = new PaymentService();
25     public static void main(String[] args) {
26         runMainMenu();
27     }
28     private static void runMainMenu() {
29         boolean exit = false;
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32             printHeader("SMART VEHICLE RENTAL MANAGEMENT SYSTEM");
33             System.out.println("1. Admin Login");
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35             System.out.println("3. Customer Login");
36             System.out.println("4. Exit");
37         }
38     }
39 }
```



SAMPLE-2:

Adding the admin details in the MONGODB compass so that only admin can access the Admin login.[1st 2 screenshots shows that]



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 - models
 - Admin.java
 - Customer.java
 - Payment.java
 - Rental.java
 - Vehicle.java
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 - AdminService.java
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 - utils
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 - InputValidator.java
 - src/main/resources
 - src/test/java
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 - JRE System Library [JavaSE-1.8]
 - Maven Dependencies
 - src
 - target
 - pom.xml

VehicleRentalSystem [Java Application] C:\Program Files\Java\jre1.8.0_491\bin\javaw.exe (18-Jun-2026, 2:14:36 PM)

```
===== SMART VEHICLE RENTAL MANAGEMENT SYSTEM =====
1. Admin Login
2. Customer Register
3. Customer Login
4. Exit
Enter your choice: 1

===== ADMIN LOGIN =====
Username: YUVA_ADMIN
Password: Yuvanesh007
Admin login successful. Welcome, YUVA_ADMIN!

===== ADMIN DASHBOARD - YUVA_ADMIN =====
1. Manage Vehicles
2. Manage Customers
3. Manage Rentals
4. Manage Payments
5. Logout
Enter your choice: 1

===== MANAGE VEHICLES =====
1. Add Vehicle
2. View All Vehicles
3. Search Vehicle by ID
4. Search Vehicles by Type
```

VehicleRentalSystem.java

```
1 package main;
2 import database.MongoConnection;
18 public class VehicleRentalSystem {
19     private static final Scanner scanner = new Scanner(System.in);
20     private static final AdminService adminService = new AdminService();
21     private static final CustomerService customerService = new CustomerService();
22 }
```

Package Explorer: SmartVehicleRentalSystem

- src/main/java
 - database
 - MongoConnection.java
 - main
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VehicleRentalSystem [Java Application] C:\Program Files\Java\jre1.8.0_491\bin\javaw.exe (18-Jun-2026, 2:14:36 PM)

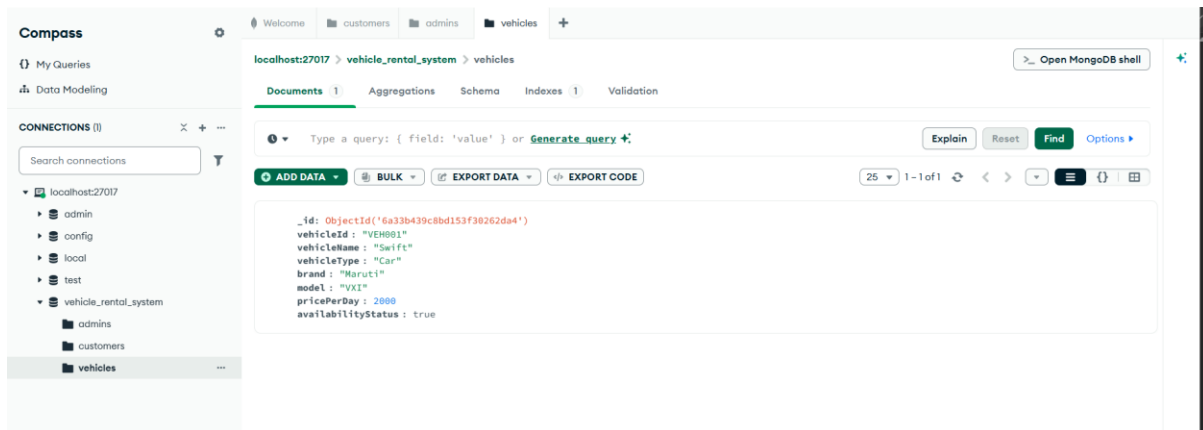
```
===== MANAGE VEHICLES =====
1. Add Vehicle
2. View All Vehicles
3. Search Vehicle by ID
4. Search Vehicles by Type
5. Search Vehicles by Brand
6. Update Vehicle
7. Delete Vehicle
8. Mark Vehicle Available/Unavailable
9. Back
Enter your choice: 1

===== ADD VEHICLE =====
Vehicle Name: Swift
Vehicle Type (e.g. Car, Bike, SUV): Car
Brand: Maruti
Model: VXi
Price Per Day: 2000
Vehicle added successfully with ID: VEH001

===== MANAGE VEHICLES =====
1. Add Vehicle
2. View All Vehicles
3. Search Vehicle by ID
4. Search Vehicles by Type
```

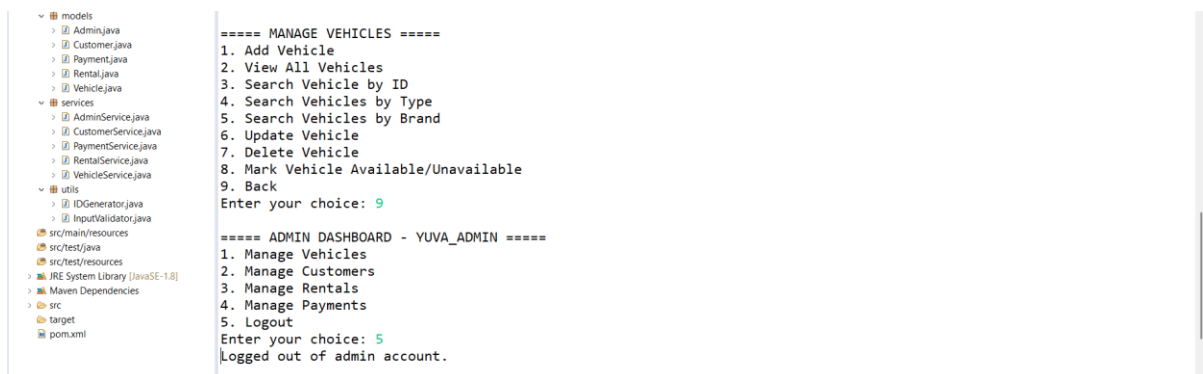
VehicleRentalSystem.java

```
1 package main;
2 import database.MongoConnection;
18 public class VehicleRentalSystem {
19     private static final Scanner scanner = new Scanner(System.in);
20     private static final AdminService adminService = new AdminService();
21     private static final CustomerService customerService = new CustomerService();
22 }
```

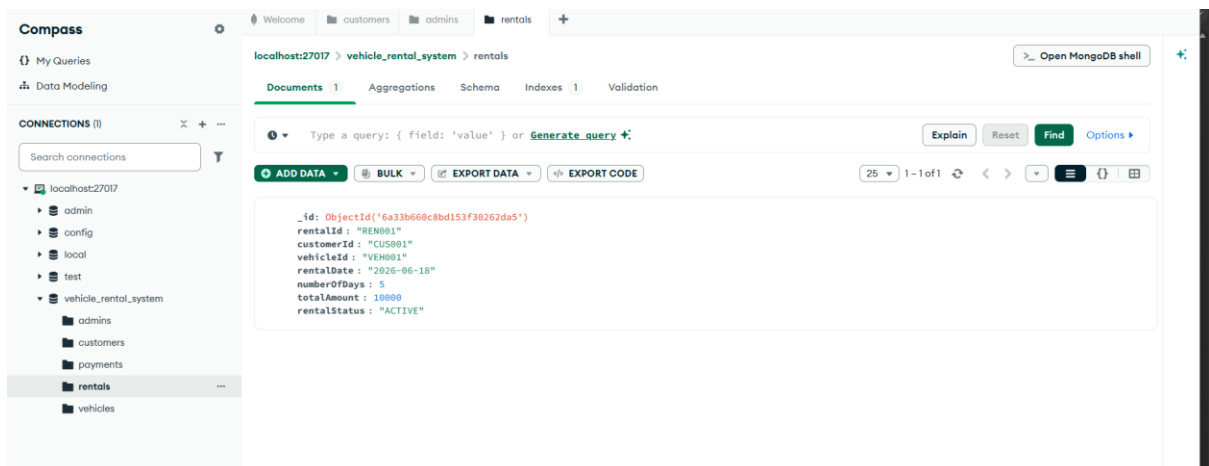
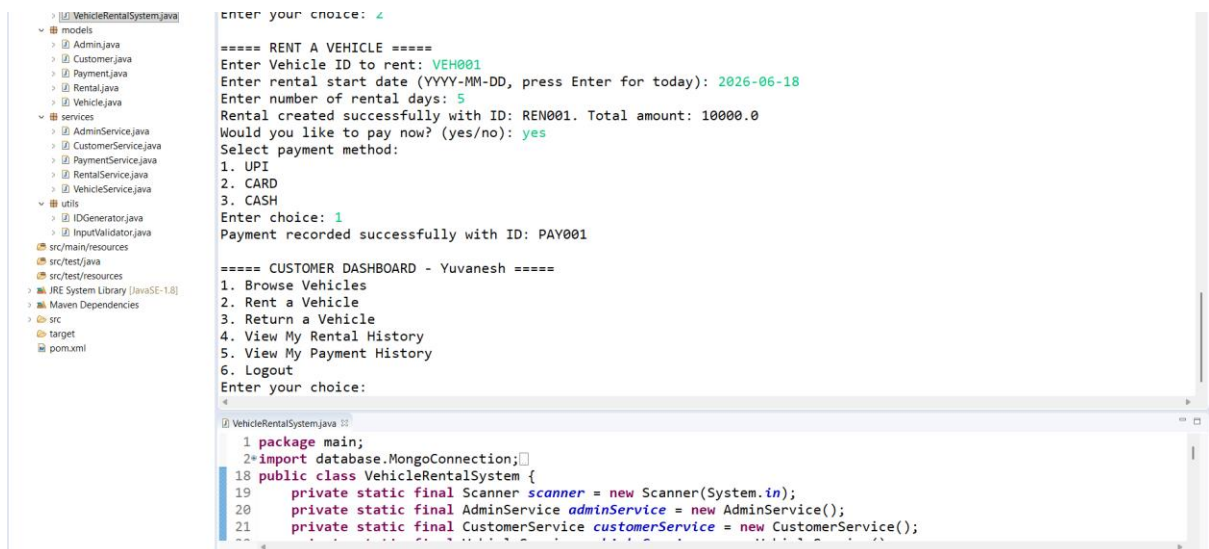
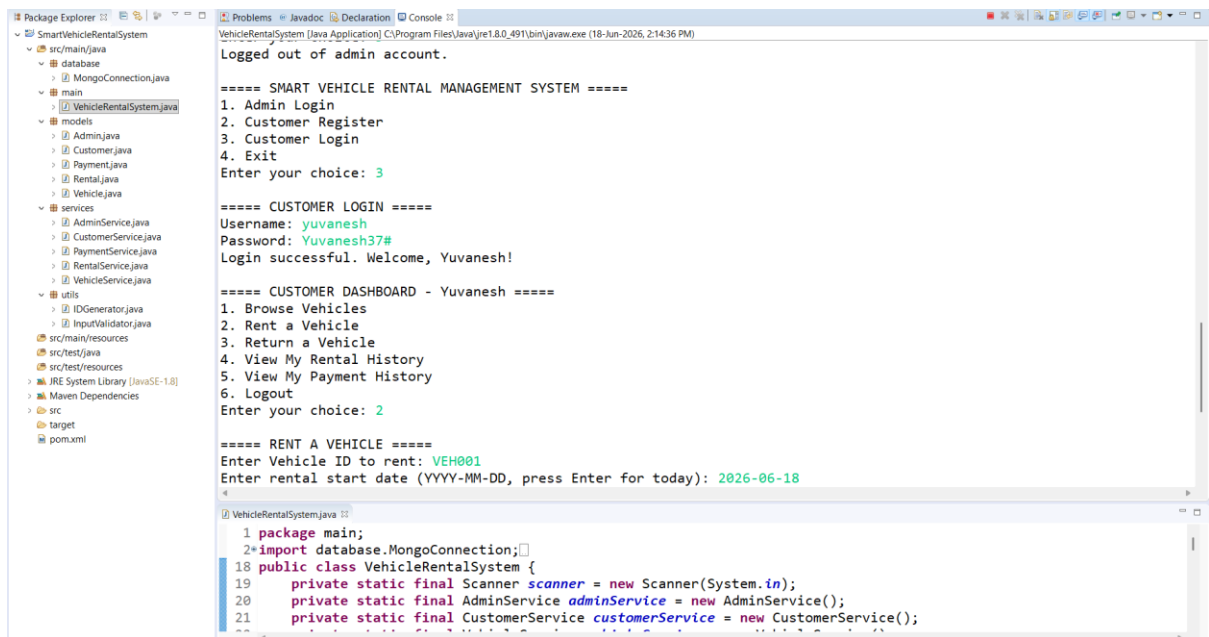


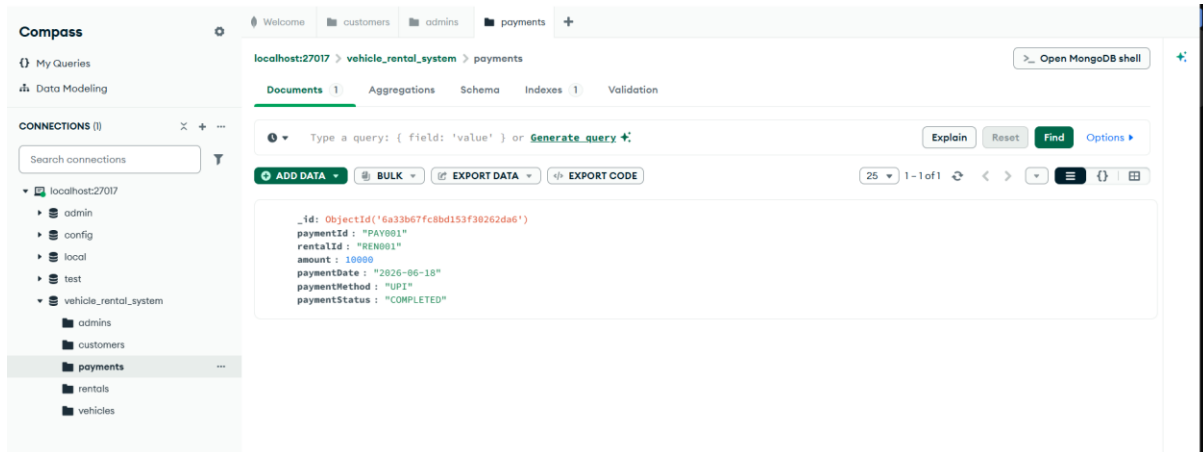
SAMPLE-3:

Before going into sample 3 we need to come back to the main interface because we log in as admin we have to logout so that screenshot also added.



NOW SAMPLE-3 STARTS...



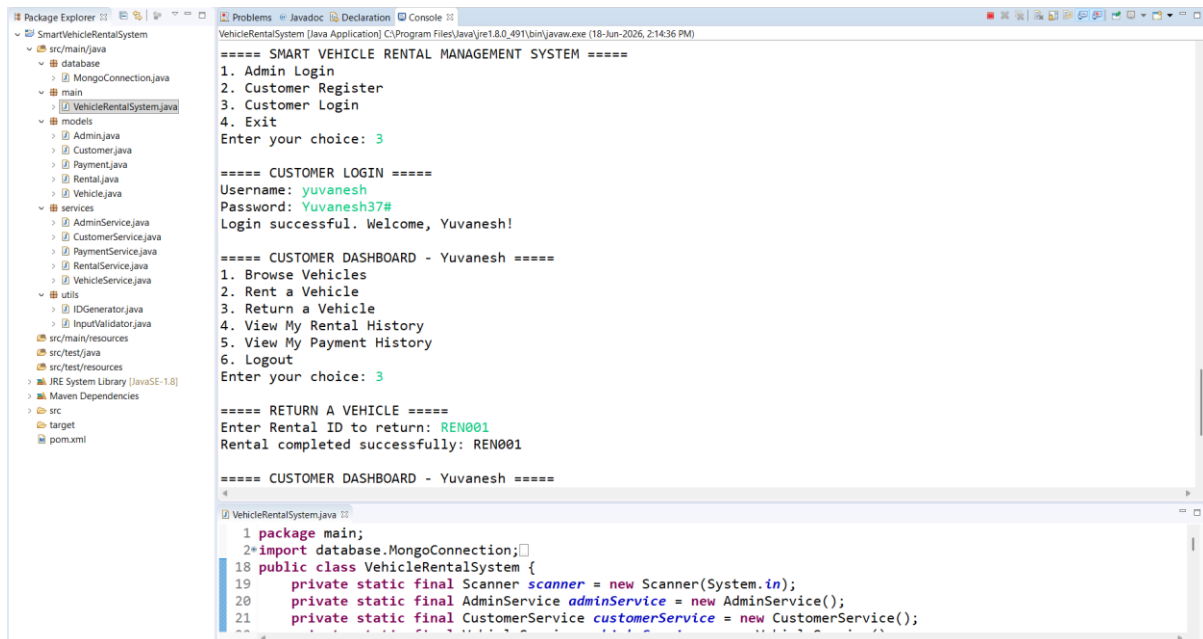


SAMPLE-4:

As we are already login as customer we need to log out so that we can test our next sample with ease.



SAMPLE-4 STARTS...



The screenshot shows an IDE with a project structure on the left and a terminal window at the bottom. The project structure includes files like Customer.java, Payment.java, Rental.java, Vehicle.java, and various service and utility classes. The terminal window displays the following output:

```
===== RETURN A VEHICLE =====
Enter Rental ID to return: REN001
Rental completed successfully: REN001

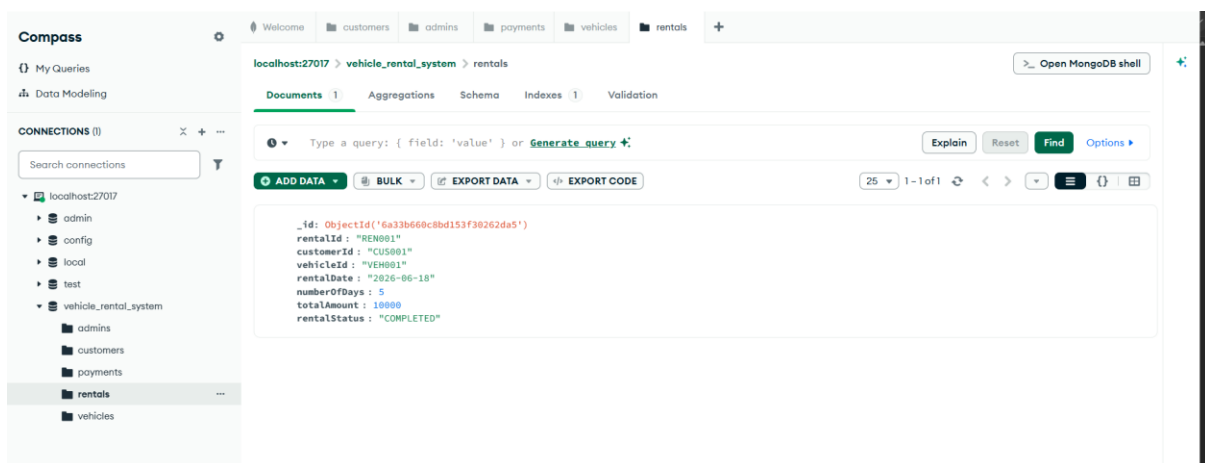
===== CUSTOMER DASHBOARD - Yuvanesh =====
1. Browse Vehicles
2. Rent a Vehicle
3. Return a Vehicle
4. View My Rental History
5. View My Payment History
6. Logout
Enter your choice: 6
Logged out of customer account.

===== SMART VEHICLE RENTAL MANAGEMENT SYSTEM =====
1. Admin Login
2. Customer Register
3. Customer Login
4. Exit
Enter your choice:
```

The code editor shows the following Java code for VehicleRentalSystem.java:

```
1 package main;
2 import database.MongoConnection;
18 public class VehicleRentalSystem {
19     private static final Scanner scanner = new Scanner(System.in);
20     private static final AdminService adminService = new AdminService();
21     private static final CustomerService customerService = new CustomerService();
22 }
```

The below screen shot is from compass before the rental status is ACTIVE now it clearly shows that vehicle has been returned by term COMPLETED.



The below screen shot is from compass before the availabilitystatus is false now it clearly shows that vehicle has been returned by term true.

